

Incompatibilities Between Fairness Metrics: A Self-Contained Guide

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Abstract

This document explains, with full notation, elementary algebra, and worked numerical examples, why the main fairness metrics used in the AI literature cannot in general hold at the same time, when exactly they conflict, and when they do not. The metrics covered are those implemented in the `fairness_metrics` toolkit (Salas-Rajo, 2026): statistical parity and disparate impact; equal opportunity and equalized odds; predictive parity (PPV and NPV parity); treatment equality; accuracy parity; the equalized disincentive (Youden) criterion; and the multidimensional subgroup metrics. The core results are the incompatibility theorems of Chouldechova (2017) and Kleinberg, Mullainathan and Raghavan (2017), together with the “independence–separation–sufficiency” organization of Barocas, Hardt and Narayanan (2023). All proofs use nothing beyond Bayes’ rule and basic probability.

Contents

1	Introduction	2
2	Setup and notation	3
3	The metrics, organized into families	4
3.1	Family 1: Independence ($\hat{Y} \perp G$)	4
3.2	Family 2: Separation ($\hat{Y} \perp G \mid Y$)	4
3.3	Family 3: Sufficiency ($Y \perp G \mid \hat{Y}$)	4
3.4	Other metrics in the toolkit	5
4	Four identities that drive everything	5
5	The incompatibility results	6
5.1	Independence vs. Separation	6
5.2	Independence vs. Sufficiency	7
5.3	Separation vs. Sufficiency: Chouldechova’s theorem	7
5.4	The score version: Kleinberg, Mullainathan and Raghavan	8
6	When the metrics are compatible	9
6.1	Equal base rates: everything can hold at once	9
6.2	Perfect prediction: families 2 and 3 reconcile, family 1 does not	10
6.3	Weakening a family restores compatibility	10
6.4	Approximate parity buys nothing structurally	10

7	The remaining binary metrics	11
7.1	Overall accuracy parity	11
7.2	Treatment equality	11
7.3	Equalized disincensive (Youden’s J)	11
8	Multidimensional metrics: new conflicts, inherited conflicts	12
8.1	An incompatibility specific to this setting: marginal vs. subgroup parity	12
8.2	Inherited conflicts, amplified	12
9	Summary	12

1 Introduction

Suppose an algorithm decides who gets a loan. A natural question is whether the algorithm is fair to a protected group (say, group a) relative to a reference group (say, group b). The AI literature has proposed many numerical answers to this question. Each of them sounds reasonable on its own:

- “Both groups should be approved at the same rate.” (statistical parity)
- “Among people who would truly repay, both groups should be approved at the same rate.” (equal opportunity)
- “Among people who are approved, the repayment rate should be the same in both groups.” (predictive parity)

The central message of this document is that these reasonable-sounding demands are, in general, *mutually incompatible*: except in special circumstances, no algorithm — however sophisticated — can satisfy all of them at once. This is not an engineering limitation. It is a mathematical fact, provable with elementary algebra, and it holds for human decision makers just as much as for algorithms.

The two special circumstances under which the conflicts disappear are worth stating up front, because they organize everything that follows:

1. **Equal base rates.** If the true outcome is equally common in both groups (e.g., the true repayment rate is the same for men and women), most conflicts vanish.
2. **Perfect prediction.** If the algorithm never makes a mistake, some (but not all) conflicts vanish.

Neither condition typically holds in practice. Base rates differ across groups in almost every applied setting — often precisely because of the upstream disadvantage that motivates the fairness audit in the first place — and no real predictor is perfect. Hence the incompatibilities bite, and choosing a fairness metric means choosing which kind of disparity one is willing to tolerate.

Section 2 sets up notation. Section 3 defines the metrics and organizes them into three families. Section 4 derives the small set of identities from which every incompatibility follows. Section 5 states and proves the incompatibility results, each followed by a worked numerical example. Section 6 collects the cases where metrics *are* compatible, which is just as important to understand.

Section 7 covers the metrics not directly involved in the classical theorems (accuracy parity, treatment equality, the Youden criterion). Section 8 extends the discussion to the multidimensional subgroup metrics. Section 9 summarizes everything in one table.

2 Setup and notation

There is a population of individuals. For each individual we observe:

- $G \in \{a, b\}$: the protected group. Group a is the disadvantaged group, group b the reference group. (In the toolkit, $G = 0$ and $G = 1$.)
- $Y \in \{0, 1\}$: the true outcome. $Y = 1$ is the “good” outcome (e.g., the person would repay the loan).
- $\hat{Y} \in \{0, 1\}$: the algorithm’s decision. $\hat{Y} = 1$ is the favorable decision (e.g., the loan is approved).
- $S \in [0, 1]$: an underlying score (e.g., predicted repayment probability). Typically $\hat{Y} = \mathbf{1}\{S \geq \tau\}$ for some threshold τ . The score matters only in Section 5.4; everywhere else, $\hat{Y}$ suffices.

The single most important quantity in this document is the **base rate** (or prevalence) of each group:

$$p_g = \Pr(Y = 1 \mid G = g), \quad g \in \{a, b\}. \quad (1)$$

In words: the fraction of group g that truly deserves the favorable outcome. Almost every incompatibility below reduces to the statement “*this combination of metrics forces $p_a = p_b$; so if $p_a \neq p_b$, the combination is impossible.*”

Within each group, every individual falls into one of four cells of the confusion matrix:

	$\hat{Y} = 1$ (approved)	$\hat{Y} = 0$ (denied)
$Y = 1$ (would repay)	true positive (TP)	false negative (FN)
$Y = 0$ (would default)	false positive (FP)	true negative (TN)

From the confusion matrix of group g we define the standard rates. Rates in the left column condition on the *truth* Y ; rates in the right column condition on the *decision* $\hat{Y}$:

Conditioning on the truth	Conditioning on the decision
$\text{TPR}_g = \Pr(\hat{Y} = 1 \mid Y = 1, G = g)$	$\text{PPV}_g = \Pr(Y = 1 \mid \hat{Y} = 1, G = g)$
$\text{FNR}_g = \Pr(\hat{Y} = 0 \mid Y = 1, G = g) = 1 - \text{TPR}_g$	$\text{NPV}_g = \Pr(Y = 0 \mid \hat{Y} = 0, G = g)$
$\text{FPR}_g = \Pr(\hat{Y} = 1 \mid Y = 0, G = g)$	
$\text{TNR}_g = \Pr(\hat{Y} = 0 \mid Y = 0, G = g) = 1 - \text{FPR}_g$	

Finally, the **acceptance rate** (or selection rate) of group g is

$$r_g = \Pr(\hat{Y} = 1 \mid G = g), \quad (2)$$

and the **accuracy** is $\text{Acc}_g = \Pr(\hat{Y} = Y \mid G = g)$.

Remark 1 (Reading the two columns). The distinction between the two columns of rates is the conceptual heart of the whole topic. TPR answers: “*given that you deserved approval, what was your chance of getting it?*” PPV answers: “*given that you were approved, what was the chance you deserved it?*” Both are reasonable things to equalize across groups. The theorems below say you usually cannot equalize both.

3 The metrics, organized into families

The thirteen binary metrics in the toolkit look heterogeneous, but they fall into three families plus a residual category. The three families correspond to three conditional-independence conditions on the pair $(\hat{Y}, G)$, an organization due to Barocas, Hardt and Narayanan (2023). Grouping the metrics this way is what makes the incompatibility results transparent: the theorems are statements about the *families*, and each metric inherits the conflicts of its family.

3.1 Family 1: Independence ($\hat{Y} \perp G$)

The decision should not depend on the group at all. Statistically: $\Pr(\hat{Y} = 1 \mid G = a) = \Pr(\hat{Y} = 1 \mid G = b)$, i.e., $r_a = r_b$.

Definition 1 (Statistical Parity, SP). $\text{SP_diff} = r_a - r_b$. Parity holds when the difference is 0.

Definition 2 (Disparate Impact, DI). $\text{DI_ratio} = r_a/r_b$. Parity holds when the ratio is 1. This is the same condition as SP written multiplicatively; U.S. employment practice uses the “four-fifths rule” ($\text{DI_ratio} \geq 0.8$) as a red flag (Feldman et al., 2015).

Note what independence does *not* look at: the truth Y . It compares raw approval rates, deservingness aside.

3.2 Family 2: Separation ($\hat{Y} \perp G \mid Y$)

The decision may depend on the truth, but — *given the truth* — not on the group. Among people who would truly repay, both groups face the same approval chance; and likewise among people who would truly default.

Definition 3 (Equalized Odds). $\text{TPR}_a = \text{TPR}_b$ and $\text{FPR}_a = \text{FPR}_b$ (Hardt, Price and Srebro, 2016). In the toolkit: $\text{EqOdds_plus_diff} = \text{TPR}_a - \text{TPR}_b$ and $\text{EqOdds_minus_diff} = \text{FPR}_a - \text{FPR}_b$, both 0 under parity.

Definition 4 (Equal Opportunity). Only the condition on the deserving: $\text{TPR}_a = \text{TPR}_b$ (EqOp_plus_diff). The mirror version equalizes TNR (EqOp_minus_diff). Equal opportunity is a deliberate *weakening* of equalized odds, and this weakening matters for compatibility (Section 6).

Conditional Equal Opportunity (recomputing the TPR gap within strata of a covariate A) belongs to this family; it refines the diagnosis but does not change any of the incompatibility logic, so we do not treat it separately.

3.3 Family 3: Sufficiency ($Y \perp G \mid \hat{Y}$)

Flip the conditioning: given the *decision*, the truth should not depend on the group. An approval should “mean the same thing” in both groups.

Definition 5 (Predictive Parity / PPV parity). $PPV_a = PPV_b$ (Chouldechova, 2017). Toolkit: `PPV_parity_diff`.

Definition 6 (NPV parity). $NPV_a = NPV_b$. Toolkit: `NPV_parity_diff`. Full sufficiency for a binary decision is PPV parity plus NPV parity.

For scores, the family-3 notion is **calibration within groups**: $\Pr(Y = 1 \mid S = s, G = g) = s$ for all s and both g . A score of 0.7 should mean a 70% repayment chance whether you are in group a or b . This is the version used in the Kleinberg–Mullainathan–Raghavan theorem (Section 5.4).

3.4 Other metrics in the toolkit

Three binary metrics do not fit neatly into the three families; they are functions of the family-2 error rates.

Definition 7 (Treatment Equality, TE). Equality of the ratio FPR_g/FNR_g across groups, in difference or ratio form (Berk et al., 2021). The idea: the *mix* of the two error types should be the same in both groups, even if error levels differ.

Definition 8 (Overall Accuracy parity). $Acc_a = Acc_b$, in difference or ratio form (Berk et al., 2021).

Definition 9 (Equalized Disincentive). Equality of Youden’s J statistic, $J_g = TPR_g - FPR_g$, across groups (group-parity version of Youden, 1950). J_g measures how much better than random the classifier is within group g .

The four multidimensional metrics (Statistical Subgroup Parity, False Positive Subgroup Parity, Differential Fairness, Worst-Case Fairness) extend family-1 and family-2 ideas from two groups to many subgroups; they are treated in Section 8.

4 Four identities that drive everything

Every incompatibility result below follows from four elementary identities. All four are exercises in Bayes’ rule and the law of total probability, applied *within* a group g (so all probabilities condition on $G = g$; we keep the subscript g on the rates and drop it from $\Pr$).

Identity 1: acceptance rate. Split the accepted into deserving and undeserving:

$$r_g = p_g TPR_g + (1 - p_g) FPR_g. \quad (3)$$

The acceptance rate is a base-rate-weighted average of the two family-2 rates. This links family 1 to family 2.

Identity 2: precision as a function of error rates. By Bayes’ rule,

$$PPV_g = \frac{p_g TPR_g}{p_g TPR_g + (1 - p_g) FPR_g}, \quad NPV_g = \frac{(1 - p_g) TNR_g}{(1 - p_g) TNR_g + p_g FNR_g}. \quad (4)$$

Family-3 quantities are determined by family-2 quantities *and the base rate*. That last part is the source of all the trouble: two groups can have identical error rates and still have different PPVs, simply because their base rates differ.

Identity 3: Chouldechova’s rearrangement. Solving (4) for FPR_g :

$$\text{FPR}_g = \frac{p_g}{1 - p_g} \cdot \frac{1 - \text{PPV}_g}{\text{PPV}_g} \cdot (1 - \text{FNR}_g). \quad (5)$$

This is the display equation of Chouldechova (2017). It ties together, in a single line, one family-2 rate on the left and the other family-2 rate, the family-3 rate, and the base rate on the right. Fix any two of (FPR, FNR, PPV) and the base rate, and the third is pinned down.

Identity 4: recovering the base rate from the decision side. Split the deserving into accepted and rejected:

$$p_g = r_g \text{PPV}_g + (1 - r_g)(1 - \text{NPV}_g). \quad (6)$$

The base rate is recoverable from the acceptance rate plus the two family-3 quantities. This links family 1 to family 3.

One more identity is useful for accuracy parity:

$$\text{Acc}_g = 1 - p_g \text{FNR}_g - (1 - p_g) \text{FPR}_g. \quad (7)$$

Remark 2 (The moral of the identities). The three families are not three independent dials. Given the base rate p_g , any two families’ worth of quantities pin down the third. Demanding parity in two families therefore imposes a constraint that, generically, only equal base rates can satisfy. The next section makes this precise, pair of families by pair of families.

5 The incompatibility results

Throughout this section, “degenerate classifier” means one that accepts everyone, rejects everyone, or whose decision carries no information about Y within a group; and “perfect classifier” means $\text{TPR}_g = 1$ and $\text{FPR}_g = 0$ in both groups.

5.1 Independence vs. Separation

Proposition 1 (SP and Equalized Odds conflict). *Suppose equalized odds holds: $\text{TPR}_a = \text{TPR}_b = \text{TPR}$ and $\text{FPR}_a = \text{FPR}_b = \text{FPR}$. Then statistical parity ($r_a = r_b$) holds if and only if*

$$(p_a - p_b)(\text{TPR} - \text{FPR}) = 0,$$

that is, if and only if base rates are equal or $\text{TPR} = \text{FPR}$ (the classifier’s decision is uninformative about Y).

Proof. By identity (3), $r_g = p_g \text{TPR} + (1 - p_g) \text{FPR} = \text{FPR} + p_g(\text{TPR} - \text{FPR})$. Hence $r_a - r_b = (p_a - p_b)(\text{TPR} - \text{FPR})$. $\square$

In words: if the classifier treats deserving and undeserving people identically *within each truth category* (equalized odds) and is at all useful ($\text{TPR} > \text{FPR}$), then the group with the higher base rate will mechanically be accepted more often. Acceptance-rate gaps are then not evidence of differential treatment; they are the arithmetic shadow of the base-rate gap.

Example 1 (Equalized odds forces an acceptance gap). Two groups of 1,000 loan applicants each. Group a has base rate $p_a = 0.2$ (200 would repay), group b has $p_b = 0.5$ (500 would repay). The algorithm satisfies equalized odds exactly, with $\text{TPR} = 0.8$ and $\text{FPR} = 0.2$ in both groups.

	Group a ($p_a = 0.2$)		Group b ($p_b = 0.5$)	
	$\hat{Y} = 1$	$\hat{Y} = 0$	$\hat{Y} = 1$	$\hat{Y} = 0$
$Y = 1$	160	40	400	100
$Y = 0$	160	640	100	400

Check group a : $\text{TPR}_a = 160/200 = 0.8$, $\text{FPR}_a = 160/800 = 0.2$. Check group b : $\text{TPR}_b = 400/500 = 0.8$, $\text{FPR}_b = 100/500 = 0.2$. Equalized odds holds. But:

$$r_a = \frac{160 + 160}{1000} = 0.32, \quad r_b = \frac{400 + 100}{1000} = 0.50.$$

So $\text{SP_diff} = -0.18$ and $\text{DI_ratio} = 0.64$, failing the four-fifths rule — even though the algorithm makes exactly the same errors, at the same rates, in both groups. And note the PPVs: $\text{PPV}_a = 160/320 = 0.5$ versus $\text{PPV}_b = 400/500 = 0.8$. Sufficiency fails too; that is the subject of Section 5.3.

5.2 Independence vs. Sufficiency

Proposition 2 (SP and full predictive parity force equal base rates). *Suppose $0 < r_g < 1$ in both groups (the classifier is not degenerate). If statistical parity holds ($r_a = r_b$), PPV parity holds, and NPV parity holds, then $p_a = p_b$.*

Proof. Identity (6) writes p_g as a function of $(r_g, \text{PPV}_g, \text{NPV}_g)$ only. If all three arguments are equal across groups, so is p_g . $\square$

The intuition is immediate: if both groups are accepted at the same rate, and acceptances are equally reliable in both groups, and rejections are equally reliable in both groups, then the two groups must contain the same fraction of deserving individuals. Contrapositive: with unequal base rates, an algorithm satisfying full sufficiency *must* produce unequal acceptance rates.

5.3 Separation vs. Sufficiency: Chouldechova’s theorem

This is the incompatibility made famous by the COMPAS recidivism debate. ProPublica criticized the COMPAS risk tool because its false positive rate was much higher for Black defendants than for white defendants (a family-2 disparity); Northpointe, the vendor, replied that the tool was equally predictive for both groups (family-3 parity). Chouldechova (2017) showed that, given the different arrest-recidivism base rates in the data, both sides were describing the same mathematical necessity: with unequal base rates, one of the two disparities *must* appear.

Proposition 3 (Chouldechova, 2017). *Suppose $p_a \neq p_b$, and suppose the classifier satisfies simultaneously*

$$\text{FNR}_a = \text{FNR}_b, \quad \text{FPR}_a = \text{FPR}_b, \quad \text{PPV}_a = \text{PPV}_b.$$

Then the classifier is degenerate: $\text{FPR} = 0$ together with either $\text{FNR} = 1$ (nobody is accepted) or $\text{PPV} = 1$. In particular, a non-degenerate, imperfect classifier cannot equalize the two error rates and the PPV across groups with unequal base rates.

Proof. Apply identity (5) in each group and equate:

$$\frac{p_a}{1 - p_a} \cdot \frac{1 - \text{PPV}}{\text{PPV}} \cdot (1 - \text{FNR}) = \text{FPR} = \frac{p_b}{1 - p_b} \cdot \frac{1 - \text{PPV}}{\text{PPV}} \cdot (1 - \text{FNR}).$$

The map $p \mapsto p/(1-p)$ is strictly increasing, so $p_a \neq p_b$ makes the two outer expressions differ *unless* the common factor $\frac{1-\text{PPV}}{\text{PPV}}(1-\text{FNR})$ is zero, i.e., $\text{PPV} = 1$ or $\text{FNR} = 1$; and in either case the middle term forces $\text{FPR} = 0$. $\square$

Remark 3 (Where the perfect classifier fits). A perfect classifier ($\text{TPR} = 1$, $\text{FPR} = 0$) has $\text{PPV} = \text{NPV} = 1$ and $\text{FNR} = \text{FPR} = 0$ in both groups, so it satisfies separation and sufficiency simultaneously even with unequal base rates — it lands exactly in the degenerate branch of the proposition ($\text{FPR} = 0$, $\text{PPV} = 1$). What it does *not* satisfy is independence: by identity (3), its acceptance rates are $r_g = p_g$, so statistical parity fails whenever base rates differ. Perfect prediction reconciles families 2 and 3, never family 1.

Example 2 (Same numbers, viewed from the other side). Return to Example 1. Equalized odds held ($\text{FNR} = \text{FPR} = 0.2$ in both groups), and we found $\text{PPV}_a = 0.5$ versus $\text{PPV}_b = 0.8$. An approval “means” an 80% repayment chance in group b but only a coin flip in group a . A lender auditing with PPV parity would call this algorithm unfair to group b ’s standards of meaning; an auditor using equalized odds would call it perfectly fair. Proposition 3 says no retraining can satisfy both auditors while base rates are 0.2 versus 0.5.

The NPVs tell the mirror story: $\text{NPV}_a = 640/680 \approx 0.94$ versus $\text{NPV}_b = 400/500 = 0.80$. Rejections are more reliable in the low-base-rate group. Nothing is broken; this is Bayes’ rule at work.

Example 3 (Equalizing PPV instead, and watching FPR break). Keep the same populations and keep $\text{TPR} = 0.8$ in both groups, but now *demand* $\text{PPV}_a = \text{PPV}_b = 0.8$. For group b nothing changes. For group a , identity (5) with $p_a = 0.2$, $\text{PPV} = 0.8$, $\text{FNR} = 0.2$ gives

$$\text{FPR}_a = \frac{0.2}{0.8} \cdot \frac{0.2}{0.8} \cdot 0.8 = 0.05.$$

	Group a ($p_a = 0.2$)		Group b ($p_b = 0.5$)	
	$\hat{Y} = 1$	$\hat{Y} = 0$	$\hat{Y} = 1$	$\hat{Y} = 0$
$Y = 1$	160	40	400	100
$Y = 0$	40	760	100	400

Now $\text{PPV}_a = 160/200 = 0.8 = \text{PPV}_b$ and $\text{TPR}_a = \text{TPR}_b = 0.8$, but $\text{FPR}_a = 40/800 = 0.05$ versus $\text{FPR}_b = 0.2$: equalized odds now fails in its FPR component, by a factor of four. The disparity did not disappear when we equalized PPV; it migrated from the PPV column to the FPR column. (Acceptance rates: $r_a = 0.2$, $r_b = 0.5$; $\text{DI_ratio} = 0.4$, so independence fails badly here as well.) This migration of disparities across columns of the confusion matrix is the practical face of Proposition 3.

5.4 The score version: Kleinberg, Mullainathan and Raghavan

Chouldechova’s theorem is about a binary decision. Kleinberg, Mullainathan and Raghavan (2017) prove a parallel impossibility one level earlier, at the level of the *score* S , and it is worth having both because many audits (including COMPAS) examine scores rather than thresholded decisions. They consider three conditions:

1. **Calibration within groups:** $\Pr(Y = 1 \mid S = s, G = g) = s$ for every score value s and both groups. (The family-3 condition for scores.)

2. **Balance for the positive class:** the average score among truly positive individuals is the same in both groups: $\mathbb{E}[S \mid Y = 1, G = a] = \mathbb{E}[S \mid Y = 1, G = b]$. (A score analogue of equal TPR.)
3. **Balance for the negative class:** $\mathbb{E}[S \mid Y = 0, G = a] = \mathbb{E}[S \mid Y = 0, G = b]$. (A score analogue of equal FPR.)

Proposition 4 (Kleinberg et al., 2017; informal statement). *A score can satisfy all three conditions simultaneously only if base rates are equal ($p_a = p_b$) or the score is perfect (every individual receives $S \in \{0, 1\}$ and $S = Y$).*

Proof sketch. The key observation is that calibration pins down the group mean of the score: taking expectations, calibration implies $\mathbb{E}[S \mid G = g] = \Pr(Y = 1 \mid G = g) = p_g$. Decompose that mean by true class:

$$p_g = \mathbb{E}[S \mid G = g] = p_g \mu_g^+ + (1 - p_g) \mu_g^-,$$

where $\mu_g^+ = \mathbb{E}[S \mid Y = 1, G = g]$ and $\mu_g^- = \mathbb{E}[S \mid Y = 0, G = g]$. Balance makes μ^+ and μ^- common across groups, so the displayed equation reads $p_g = \mu^- + p_g(\mu^+ - \mu^-)$ for both values of g . Subtracting the two group versions: $(p_a - p_b)(1 - (\mu^+ - \mu^-)) = 0$. So either $p_a = p_b$, or $\mu^+ - \mu^- = 1$, which for scores in $[0, 1]$ forces $\mu^+ = 1$ and $\mu^- = 0$: positives all scored 1, negatives all scored 0 — perfect prediction. (The full paper proves an approximate version too: if the three conditions hold approximately, the setting must be approximately one of the two special cases. The conflict is therefore not a knife-edge artifact; being “close to fair” on all three requires being close to equal base rates or close to perfection.) $\square$

Remark 4 (How the two theorems relate). Both theorems say: family 3 plus family 2 forces equal base rates or perfection. Chouldechova’s version is exact algebra on the thresholded confusion matrix; KMR’s version works on the underlying score and adds the robustness result for approximate fairness. For the purposes of the toolkit’s metrics, Proposition 3 is the directly relevant statement, with Proposition 4 as its score-level backbone. A useful companion is Pleiss et al. (2017), who show that even *relaxed* versions of equalized odds are compatible with calibration only through trivial devices (e.g., randomly withholding predictions for some individuals), reinforcing that the tension is structural.

6 When the metrics are compatible

The impossibility results are often summarized as “fairness metrics are incompatible,” which overstates them. Each theorem has explicit escape hatches, and several metric pairs never conflict at all. Knowing these cases matters as much as knowing the conflicts, because it tells you what an audit that reports many metrics can and cannot reveal.

6.1 Equal base rates: everything can hold at once

If $p_a = p_b$, every proposition above becomes vacuous. A classifier that behaves identically in both groups (same TPR, same FPR) then automatically satisfies independence (identity 3 gives $r_a = r_b$), separation, sufficiency (identity 4 gives equal PPV and NPV), accuracy parity, treatment equality, and the Youden criterion, all simultaneously.

Example 4. Give both groups of Example 1 base rate 0.5 and the same rates $\text{TPR} = 0.8$, $\text{FPR} = 0.2$. Both groups then share the confusion matrix of group b , and all thirteen binary metrics sit exactly at their parity values (0 for differences, 1 for ratios).

This is why the base-rate gap is the first thing to inspect in any audit: it is the single statistic that determines whether “fair on every metric” is even a coherent target.

6.2 Perfect prediction: families 2 and 3 reconcile, family 1 does not

A perfect classifier satisfies separation and sufficiency with any base rates, but its acceptance rate in each group equals that group’s base rate, so independence fails whenever $p_a \neq p_b$. There is no perfection escape hatch for statistical parity. Conversely, a fully degenerate classifier (accept everyone, or accept at random at a common rate) satisfies independence and separation ($\text{TPR} = \text{FPR}$) but is useless. The pattern: with unequal base rates, you can have *fair and accurate* in the family-2/3 sense, or *equal and useless*, but not everything.

6.3 Weakening a family restores compatibility

The theorems require *full* parity within a family. Weakening one side often dissolves the conflict:

- **Equal opportunity + predictive parity is generically feasible.** Chouldechova’s theorem needs *both* error rates plus PPV. Drop the FPR requirement and the conflict disappears: Example 3 exhibits a classifier with equal TPR (0.8) and equal PPV (0.8) across groups with base rates 0.2 and 0.5. The FPR absorbs the base-rate gap. This is a principled reason why Hardt, Price and Srebro (2016) proposed equal opportunity as the practical criterion: it constrains treatment of the deserving while leaving one degree of freedom open.
- **Equal opportunity + statistical parity is generically feasible** by the same one-degree-of-freedom logic: with TPR fixed and common, identity (3) lets you choose FPR_g group by group to equalize r_g , provided the required FPR_g lands in $[0, 1]$ — a feasibility condition, not an impossibility.
- **PPV parity alone, or NPV parity alone**, conflicts with nothing in family 1: Proposition 2 needs both.

The general principle: the impossibility theorems live at the level of *families*, and each family must be imposed in full for the conflict to be forced. Any single metric from one family can typically coexist with any single metric from another; it is the complete packages that collide.

6.4 Approximate parity buys nothing structurally

One might hope that demanding each metric only approximately (gaps below some tolerance) escapes the theorems. The approximate version of Proposition 4 closes this route: small violations of all conditions simultaneously are possible only when base rates are nearly equal or prediction is nearly perfect. Tolerances trade off continuously against the base-rate gap; they do not remove the trade-off.

7 The remaining binary metrics

7.1 Overall accuracy parity

Accuracy parity looks innocuous but interacts with the families through identity (7).

Proposition 5. *Under equalized odds (common FNR, common FPR), accuracy parity holds if and only if $p_a = p_b$ or $\text{FNR} = \text{FPR}$.*

Proof. $\text{Acc}_a - \text{Acc}_b = -(p_a - p_b)\text{FNR} + (p_a - p_b)\text{FPR} = (p_a - p_b)(\text{FPR} - \text{FNR})$ by (7). $\square$

So equalized odds and accuracy parity are compatible with unequal base rates precisely when the two error rates are equal to each other — as in Examples 1 ($\text{FNR} = \text{FPR} = 0.2$, and indeed $\text{Acc}_a = \text{Acc}_b = 0.8$ there). Tilt the error mix and accuracy parity breaks: with $\text{FNR} = 0.1$, $\text{FPR} = 0.3$ common to both groups, the base rates 0.2 and 0.5 give $\text{Acc}_a = 1 - 0.02 - 0.24 = 0.74$ versus $\text{Acc}_b = 1 - 0.05 - 0.15 = 0.80$.

A caution in the other direction: accuracy parity *alone* is a weak requirement. Groups can have identical accuracy with entirely different error compositions (one group’s errors all false positives, the other’s all false negatives), which matters when the two error types have asymmetric costs for the people involved.

7.2 Treatment equality

Treatment equality equalizes the ratio $\text{FPR}_g/\text{FNR}_g$. Two observations place it in the map:

- **It is implied by equalized odds.** If both FPR and FNR are common across groups, so is their ratio. It is strictly weaker: the ratios can match while the levels differ (e.g., group a with $(\text{FPR}, \text{FNR}) = (0.10, 0.05)$ and group b with $(0.30, 0.15)$ both have ratio 2).
- **As a family-2 object, it inherits the family conflicts only partially.** Because it constrains one dimension rather than two, treatment equality behaves like the “weakened” criteria of Section 6: it is generically compatible with predictive parity or with statistical parity, since a whole one-parameter family of $(\text{FPR}_g, \text{FNR}_g)$ pairs is consistent with a given ratio, leaving room to satisfy one additional cross-family constraint.

Berk et al. (2021) advance it as capturing whether the classifier trades off the two error types the same way for both groups — a question none of the three canonical families asks directly.

7.3 Equalized disincentive (Youden’s J)

$J_g = \text{TPR}_g - \text{FPR}_g$ measures within-group informativeness. Like treatment equality it is a one-dimensional functional of the family-2 pair, implied by equalized odds and strictly weaker than it. Two useful connections:

- By identity (3), $r_g = \text{FPR}_g + p_g J_g$. So under J -parity ($J_a = J_b = J$), statistical parity requires $\text{FPR}_a - \text{FPR}_b = -(p_a - p_b)J$: attainable, but only by *deliberately* giving the higher-base-rate group a lower FPR. Independence and equal informativeness can coexist, at the price of a compensating error-rate wedge.
- J -parity with $J > 0$ rules out the “useless classifier” escape hatch of Proposition 1 in both groups at once, which is why demanding informativeness plus full equalized odds plus statistical parity brings back the hard conflict.

8 Multidimensional metrics: new conflicts, inherited conflicts

The four multidimensional metrics replace the two-group comparison with a comparison across M subgroups $zg_1, \dots, zg_M$ (e.g., all combinations of sex and region). Statistical Subgroup Parity and Differential Fairness / Worst-Case Fairness are family-1 objects at subgroup level (they compare acceptance rates); False Positive Subgroup Parity is a family-2 object (it compares FPRs). Two distinct issues arise.

8.1 An incompatibility specific to this setting: marginal vs. subgroup parity

Kearns et al. (2018) call it *fairness gerrymandering*: a classifier can satisfy statistical parity on every marginal attribute while violating it grossly on the intersections.

Example 5 (Gerrymandering). Population of 1,000: two attributes, $sex \in \{M, F\}$ and $region \in \{0, 1\}$, independent and balanced, so each of the four cells has 250 people. The classifier accepts everyone in $(M, 0)$ and $(F, 1)$, and no one else. Then

$$\Pr(\hat{Y} = 1 \mid M) = \Pr(\hat{Y} = 1 \mid F) = \Pr(\hat{Y} = 1 \mid \text{region } 0) = \Pr(\hat{Y} = 1 \mid \text{region } 1) = 0.5 :$$

perfect statistical parity by sex and by region, separately. Yet at subgroup level the acceptance rates are 1, 0, 0, 1. The toolkit’s metrics detect this: Statistical Subgroup Parity with $\varepsilon = 0$ evaluates $\max_m P(zg_m) |0.5 - r_m| = 0.25 \times 0.5 = 0.125 > 0$ (unfair), and Differential Fairness / Worst-Case Fairness blow up because $\min_m \Pr(\hat{Y} = 1 \mid zg_m) = 0$.

The general point: *marginal parity and subgroup parity are different criteria, and the first does not imply the second*. This is an incompatibility of a different kind from Sections 5.1–5.4: it is not that two desiderata cannot jointly hold, but that the coarse criterion fails to certify the fine one. Auditing only marginals leaves room — adversarial or accidental — for intersectional harm. (The reverse direction is clean: exact subgroup parity across a partition implies parity on every marginal built from it, since marginal rates are averages of subgroup rates.)

8.2 Inherited conflicts, amplified

Because the subgroup metrics are family-1 and family-2 objects, every two-group theorem applies pairwise to any two subgroups with different base rates. With M subgroups the situation is strictly harder: subgroup-level acceptance parity (family 1) and subgroup-level FPR parity (family 2) can coexist only if, for every pair of subgroups, the conditions of Proposition 1 hold — effectively requiring equal base rates across *all* subgroups, an even less plausible condition than in the two-group case. Intersectionality multiplies the number of base rates that would have to coincide.

A practical remark, distinct from the mathematics: as M grows, subgroups shrink, and empirical rates become noisy; the ε tolerance in these metrics is doing double duty as a fairness relaxation and as a statistical buffer. That is an estimation problem, not an impossibility result, but in applications it binds sooner.

9 Summary

Table 1 condenses the results. “ $p_a = p_b$ ” means the combination forces equal base rates (so it is impossible whenever base rates differ); “perfect” means perfect prediction is the other escape

Table 1: When can the pair hold simultaneously (non-degenerate classifier)?

Pair of criteria	Compatible when	Source
Statistical parity + equalized odds	$p_a = p_b$, or uninformative classifier (TPR = FPR)	Prop. 1; Barocas et al. (2023)
Statistical parity + PPV & NPV parity	$p_a = p_b$ only	Prop. 2; Barocas et al. (2023)
Equalized odds + PPV parity	$p_a = p_b$, or perfect prediction	Prop. 3; Chouldechova (2017)
Calibration + both balance conditions (scores)	$p_a = p_b$, or perfect prediction; approximate versions likewise	Prop. 4; Kleinberg et al. (2017)
Equal opportunity + PPV parity	generic (FPR absorbs the gap)	Sec. 6; Ex. 3
Equal opportunity + statistical parity	generic (choose group FPRs)	Sec. 6
Equalized odds + accuracy parity	$p_a = p_b$, or FNR = FPR	Prop. 5
Equalized odds $\Rightarrow$ treatment equality, J -parity, accuracy parity if FNR = FPR	implication, not conflict	Sec. 7
Marginal SP + subgroup SP	marginal does not imply subgroup; subgroup implies marginal	Kearns et al. (2018); Ex. 5
Subgroup SP + subgroup FPR parity	equal base rates across all subgroups	Sec. 8

hatch; “generic” means the pair is compatible in general, subject only to feasibility of rates lying in $[0, 1]$.

Three take-aways.

First, the incompatibilities are driven by one scalar per group: the base rate. Every hard conflict in Table 1 is a restatement of “these criteria jointly identify the base rates, so they cannot all hold when base rates differ.” The base-rate gap is therefore the first diagnostic in any audit: it determines which parity combinations are even feasible before any model is examined.

Second, the theorems bind at the level of complete families, not individual metrics. Full separation clashes with full sufficiency; equal opportunity, treatment equality, or J -parity — each a one-dimensional slice of family 2 — coexist generically with metrics from the other families. Reading a table of thirteen metrics, one should expect blocks of them to fail together for purely arithmetic reasons (as in Examples 1–3), and should resist interpreting each failed row as an independent finding of unfairness.

Third, none of the mathematics says which family is the right one. That choice depends on what one takes the decision to owe to whom — equal treatment of the deserving, equal meaning of the decision, or equal exposure to the favorable outcome — and with unequal base rates it is a genuine choice: satisfying one family’s demand *produces* the disparity the other family measures. The theorems do not settle the normative question; they establish that it cannot be avoided.

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